

Perspective Transformation - Python OpenCV

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In **Perspective Transformation**, we can change the perspective of a given image or video for getting better insights into the required information. In Perspective Transformation, we need to provide the points on the image from which want to gather information by changing the perspective. We also need to provide the points inside which we want to display our image. Then, we get the perspective transform from the two given sets of points and wrap it with the original image.

We use **cv2.getPerspectiveTransform** and then **cv2.warpPerspective** .

cv2.getPerspectiveTransform method

Syntax: *cv2.getPerspectiveTransform(src, dst)*

Parameters:

- **src:** *Coordinates of quadrangle vertices in the source image.*
- **dst:** *Coordinates of the corresponding quadrangle vertices in the destination image.*

cv2.warpPerspective method

Syntax: *cv2.warpPerspective(src, dst, dsize)*

Parameters:

- **src:** *Source Image*
- **dst:** *output image that has the size dsize and the same type*

as src.

- **dsiz**: size of output image

Below is the Python code explaining Perspective Transformation:

```
# import necessary libraries

import cv2
import numpy as np

# Turn on Laptop's webcam
cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    # Locate points of the documents
    # or object which you want to transform
    pts1 = np.float32([[0, 260], [640, 260],
                       [0, 400], [640, 400]])
    pts2 = np.float32([[0, 0], [400, 0],
                       [0, 640], [400, 640]])

    # Apply Perspective Transform Algorithm
    matrix = cv2.getPerspectiveTransform(pts1, pts2)
    result = cv2.warpPerspective(frame, matrix, (500, 600))

    # Wrap the transformed image
    cv2.imshow('frame', frame) # Initial Capture
    cv2.imshow('frame1', result) # Transformed Capture

    if cv2.waitKey(24) == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

Output:



